SIMATS Online Programming Lab

DS PROGRAMMING

YUGITHA B
192424102RD

QUESTIONS 10 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K (0 ≤ K < N), show original→rotated→reversal steps visually, compute the sum difference, and support multiple rotations.
Sample Input/Output:
Enter 10 numbers: 1 2 3 4 5 6 7 8 9 10
Rotate Left (L)/Right (R): L
By positions: 3
Original: [1 2 3 4 5 6 7 8 9 10]
Step1: Reverse 0-2 → [3 2 1 4 5 6 7 8 9 10]
Step2: Reverse 3-9 → [3 2 1 10 9 8 7 6 5 4]
Step3: Reverse all → [4 5 6 7 8 9 10 1 2 3]
Sum difference: 0"

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
void reverse(int arr[], int start, int end)
{
  while(start<end)
  {
    int temp = arr[start];
    arr[start] = arr[end];
    arr[end] = temp;
    start++;
    end--;
  }
}
void printArray(int arr[], int n)
{
  int i;
  for(i=0; i<n ; i++)
    print("%d", arr[i]);
  print("\n");
}
int main()
{
```

OUTPUT

Your OUTPUT will appear here!

INPUT

1 2 3 4 5 6 7 8 9 10